Root Cause Analysis of Machining Chatter in CNC Milling via Physics Simulation and Causal Inference

Dang Hai Dang · RCA-for-Chatter-Simulation project · 2026

1. One-minute summary

The problem. A CNC machine suddenly starts to vibrate violently while cutting (*chatter*) and the part surface is ruined. Sensors only say “it is vibrating”, not **why**: a loose fixture, a worn tool, too deep a cut, or a bad spindle speed? Finding the real reason is called **RCA** (Root Cause Analysis).

The system. The cutting process is a **delay differential equation (DDE)**. Inputs are material–tool–machine parameters (slowly drifting), operating parameters (controllable) and uncertainties; outputs are vibration and cutting force, but on a real machine they are seen only **indirectly** through sensors such as accelerometers, microphones and spindle current.

The project has two experiments. (1) Physics simulation: stability lobes, time-domain vibration, a small Bayesian-network diagnosis. **(2) RCA benchmark:** 14 variables from an algebraic model, injected faults, 6 algorithms guessing the cause (2,500 trials per algorithm).

Main results. (1) The Bayesian network is right on the first guess **77%** of the time and within the top 3 **100%** (120 runs). (2) With 5 anomalous samples, BRCD is right first time **71%** of the time, the other causal methods ≈ 63–64%, BARO 38%, SimpleRCA 3%; from 50 samples on, the causal methods reach ≈ 99–100%.

Important. All results are on **simulated data**. In particular, the Experiment 2 benchmark feeds **quantities that cannot be measured on a real machine** (stiffness k, damping ζ, cutting coefficient K_t…) to the algorithms as observed data, so the problem is easier than reality and near-100% accuracy cannot be extrapolated to a real machine (see Sections 3.3 and 6.2).

2. What is chatter and why is its cause hard to find?

As the tool cuts, it vibrates slightly and leaves a wavy surface. The next pass meets that wavy surface and vibrates even more, leaving bigger waves… This self-amplifying loop is **regenerative chatter**. It occurs when the depth of cut a exceeds a **stability limit** a_{lim}, which depends on the stiffness and damping of the machine–tool–workpiece system, the cutting-force coefficient and the spindle speed [1].

The difficulty is the **propagation chain**: a cause lowers a_{lim} → chatter → vibration, forces, power and roughness all rise. On the sensors *everything* looks abnormal, and the quantities that change most (peak force, roughness) are **symptoms**, not causes.

3. System model of the problem

Figure 1 summarises the whole system: inputs pass through the cutting process (a DDE) and produce physical quantities; we observe them only through sensors, and the RCA algorithm must reason backwards to the cause. The subsections below answer in turn: what the inputs are, what the system is, what is observable, whether there is closed-loop control, when anomalies occur and where the “root cause” lies.

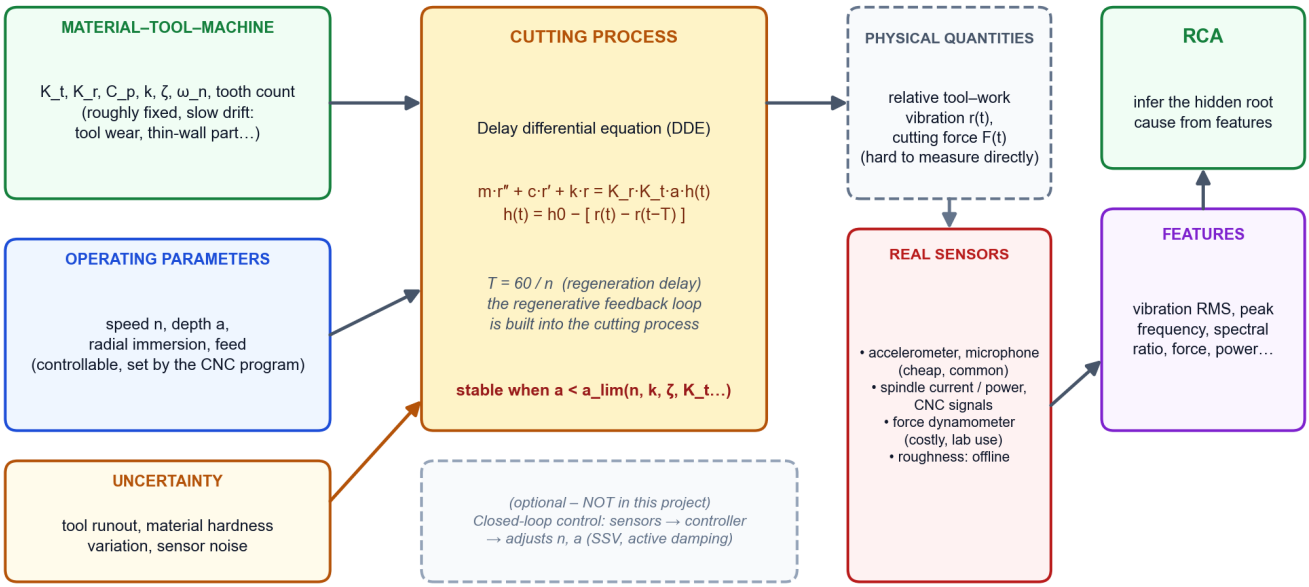


Figure 1. *Block diagram of the system. Dashed outline: not directly measurable. The closed control loop (grey dashed line at the bottom) exists in practice but is* **not** *modelled in this project.*

3.1 Inputs: three groups

Splitting inputs into “static” and “dynamic” is the right direction but not enough, because the “static” group actually **drifts over time**. Three groups are more accurate:

Table 1. *Three groups of inputs to the cutting system.*

Group	Examples	How it changes	Role in RCA
Material–tool–machine parameters	cutting coefficients K_t, K_r; process-damping coefficient C_p; stiffness k, damping ζ, natural frequency ω_n; tooth count	roughly fixed but **drifting slowly**: tool wear, thin-wall part losing material, machine axis position [1]	root-cause candidates
Operating parameters	spindle speed n, axial depth a, radial immersion, feed	controllable, set in the CNC program	candidates (bad programming, overrides)
Uncertainty	tool runout, material hardness variation, sensor noise, initial conditions	random	adds noise; can trigger chatter

3.2 The system: a delay differential equation

As you suspected, the system is a DDE. For orthogonal cutting [1, Eq. 15–16], a tool with stiffness k, damping ζ and natural frequency ω_n is driven by a force that depends on chip thickness $h(t) = h_0 - [r(t) - r(t - T)]$, where $T = 60/n$ is the regeneration delay (one revolution) and C_p is the process-damping coefficient:

$$r'' + 2\zeta\omega_n r' + \omega_n^2 r = (\omega_n^2/k)\cdot[\, K_r K_t a\cdot h(t) - aC_p r'/v_c \,]$$

The paper also gives the frequency-domain stability condition used to draw stability-lobe diagrams (Section 4.1): $a_{lim} = -1/(2K_r K_t G(\omega_c))$ [1, Eq. 19], and Tlusty’s lower bound $a_{min} = 2k\zeta/(K_r K_t)$ [1, Eq. 20]. Together they say: **stiffer and more damped → deeper cuts are possible; a larger cutting coefficient (worn tool, harder material) → shallower cuts are needed**.

Comparison with the project: the Experiment 2 surrogate has a_{lim} proportional to $k\cdot\zeta/K_t$, the **same dependence** as Eq. 20, so it is plausible in trend. However the constant 0.85 and the lobing factor $1 - 0.35\cdot\sin(4\pi\cdot\text{rpm})$ are the project’s own choices, not from the paper; the model also omits process damping and K_r. The Experiment 1 time-domain simulator is a single-mode DDE whose force drops to zero when the tool “jumps out” of the cut, with no process damping.

3.3 What can be observed on a real machine?

You are right: the quantity that decides “chatter or not” is the relative vibration between tool and workpiece, but measuring it directly at the tool tip is hard or expensive. Chatter-detection research mainly uses accelerometers, microphones and dynamometers [2]; spindle current or internal CNC signals can also carry the chatter frequency without extra sensors [3].

Table 2. *Which quantities are observable in simulation versus on a real machine.*

Quantity	In simulation	On a real machine
Relative tool–workpiece vibration r(t)	available	hard to measure directly; inferred via acceleration, sound
Acceleration / sound (vibration RMS, peak frequency, spectral ratio)	available	**observable**, cheap, common [2]
Spindle current / power, CNC signals	available	**observable**, no external sensor needed [3]
Cutting force (dynamometer)	available	accurate but costly, mostly in laboratories [2]
Surface roughness Ra	available	usually measured after cutting, not a real-time signal
Commanded a, n (programmed values)	available	**known** from the CNC program (actual values may differ due to overrides)
k, ζ, K_t, C_p, a_{lim}, chatter severity	available (known exactly)	**not directly measurable**; only inferred or measured offline (e.g. impact test for the FRF [1])

Consequence for the project. Experiment 1 follows the right principle: causes are hidden variables and the evidence is binary symptoms derived from sensor features. Experiment 2 does **not**: its 14-column table (Figure 5) treats k, ζ, K_t, a_{lim} and chatter severity as observed columns (plus sensor noise). When a cause is faulted, its own column shifts immediately, so an algorithm only has to find the shifted column (adjusted for the graph) instead of doing hidden-variable inference. This is the main reason accuracy approaches 100% and the biggest limitation of the benchmark (Section 6.2).

3.4 Open loop or closed loop?

In the project every operating input is **fixed** (each scenario has one constant pair n, a): that is **open loop**. Two kinds of “closed loop” must be distinguished:

- The **regenerative feedback loop** is built into the cutting process itself: the paper [1] notes that Merritt expressed it as a closed-loop system with delayed feedback. It is a physical property, not a controller, and the DDE in Section 3.2 already contains it.
- **Closed-loop chatter control** does exist in practice: adaptive spindle speed variation (SSV) using acceleration as feedback [4], and active damping with piezoelectric actuators [5]. Reviews report that SSV works against period-doubling chatter but not against quasi-periodic chatter above the Hopf lobe. The project does **not** model these controllers.

In my own reasoning (no direct source yet): if the machine has adaptive control, the controller compensates and may mask the root cause, and it creates cycles in the causal graph; RCA would then need different handling than in the open-loop case.

3.5 When do anomalies occur?

The system runs normally when the operating point (n, a) lies in the stable region and the parameters are nominal. But “no noise” does **not** mean “always normal”. The main causes of anomalies:

- **Wrong operating point from the start:** if $a > a_{lim}$ with nominal parameters (programmed too deep, speed in a valley), chatter is deterministic; noise only acts as a trigger.
- **Drifting system parameters** (more common than “operating inputs got changed”): cutting and process-damping coefficients change with tool wear, thin-wall workpiece dynamics change as material is removed, low-frequency modes of large machines shift with position [1, Sec. 6]. The paper concludes that prediction accuracy is still limited by measurement uncertainty, nonlinearities and time-varying dynamics.

- **Bistable zones:** below the stability boundary there are “unsafe zones” where stable cutting and chatter coexist; small disturbances decay, large enough ones switch the process to chatter [6]. The paper [1, Sec. 3.3] also describes other bifurcations (secondary Hopf, period-2, period-3). The project does not model this.
- **Forced vibration:** Arnold distinguishes forced from self-induced vibration [1]. In milling, “normal” still includes vibration at the tooth-passing frequency; it is not zero vibration.

3.6 Where does the “root cause” lie?

The parameters in the model are the **direct mechanical causes**. The root cause in the RCA sense usually lies one level deeper. The full causal chain is:

Level	Example	Observable?
1. Fault event (the real root cause)	loose clamp, long overhang, flank wear, over-deep program, wrong speed	no
2. Changed parameter (mechanical cause)	k↓, ζ↓, K_t↑, C_p, a↑, n off	no (except a, n)
3. Stability limit a_{lim} drops	$a > a_{lim}$	no
4. Chatter	growing regenerative vibration	indirectly
5. Symptoms	vibration RMS, peak frequency, force, power, roughness	yes (through sensors)

The mapping between levels 1 and 2 is **not one-to-one**: a long overhang changes both k and ζ, tool wear changes both K_t and C_p [1]. Two different faults can therefore produce similar symptoms, and RCA may only be able to separate them down to the parameter level rather than the fault-event level. **The project currently merges levels 1 and 2:** its “5 root causes” are really 5 parameters.

3.7 What has research achieved on root causes so far?

I searched the sources listed in the references. In short: the relation **parameter → stability** is very well known; the relation **condition or fault → parameter** has been measured piecemeal for individual factors; but a unified map **fault → parameter → stability → symptoms** with the sign of each effect I **did not find** in the sources I checked.

- **Firmly known:** the Tlusty and Tobias stability laws from the 1950s; the lower bound $a_{min} = 2k\zeta/(K_r K_t)$; the speed dependence through the delay $T = 60/n$; process damping adding stability at low speed [1].
- **Measured factor by factor:** a longer overhang from 35 to 65 mm drops the first natural frequency from 565 Hz to 325 Hz [15], and from 35 to 70 mm lowers a_{lim} for all three tested materials [14]; cutting coefficients vary strongly with material [14]; the dynamics of thin-wall parts and large machines vary with position [1][20].
- **Much remains uncertain:** K_t and C_p carry large uncertainty; wear changes the flank contact and hence process damping [1, p. 17]. Wear raises cutting force but also raises process damping [16][21], so the **net effect on chatter is not monotonic**.
- **Fault catalogues:** FMEA/FMECA of CNC machines lists failure modes by component (spindle wear or seizure, tool chipping or wear, clamping faults…) [18]; a knowledge-graph work for CNC fault diagnosis uses the entity types equipment, module, parameter, alarm, phenomenon, cause, solution [19]; Wehner et al. combine a causal Bayesian network with a KG for manufacturing RCA [7]. These describe faults at component or phenomenon level and do **not** quantify the effect on a_{lim}.
- **Gap:** there is no unified table or graph linking causes to machine–tool–operating parameters with signs of effect. For “loose workholding” and “worn spindle bearings” I only found engineering blogs [17], a low-reliability source.

3.8 Root causes and their relations to the parameters

Figure 2 is a graph I **compiled from the sources above**; it is not an experimental result of this project. The left column holds 9 root causes (grouped as tool–holder, workpiece, machine, operation); the middle column holds machine–tool–material parameters (yellow) and operating parameters (blue); the right column is stability. Each edge carries the **source number** from the reference list; the **line style** shows the level of verification (solid: I read it in the source; dashed: secondary source or summary only; dotted: weak source); the **colour** shows the direction of the effect on stability.

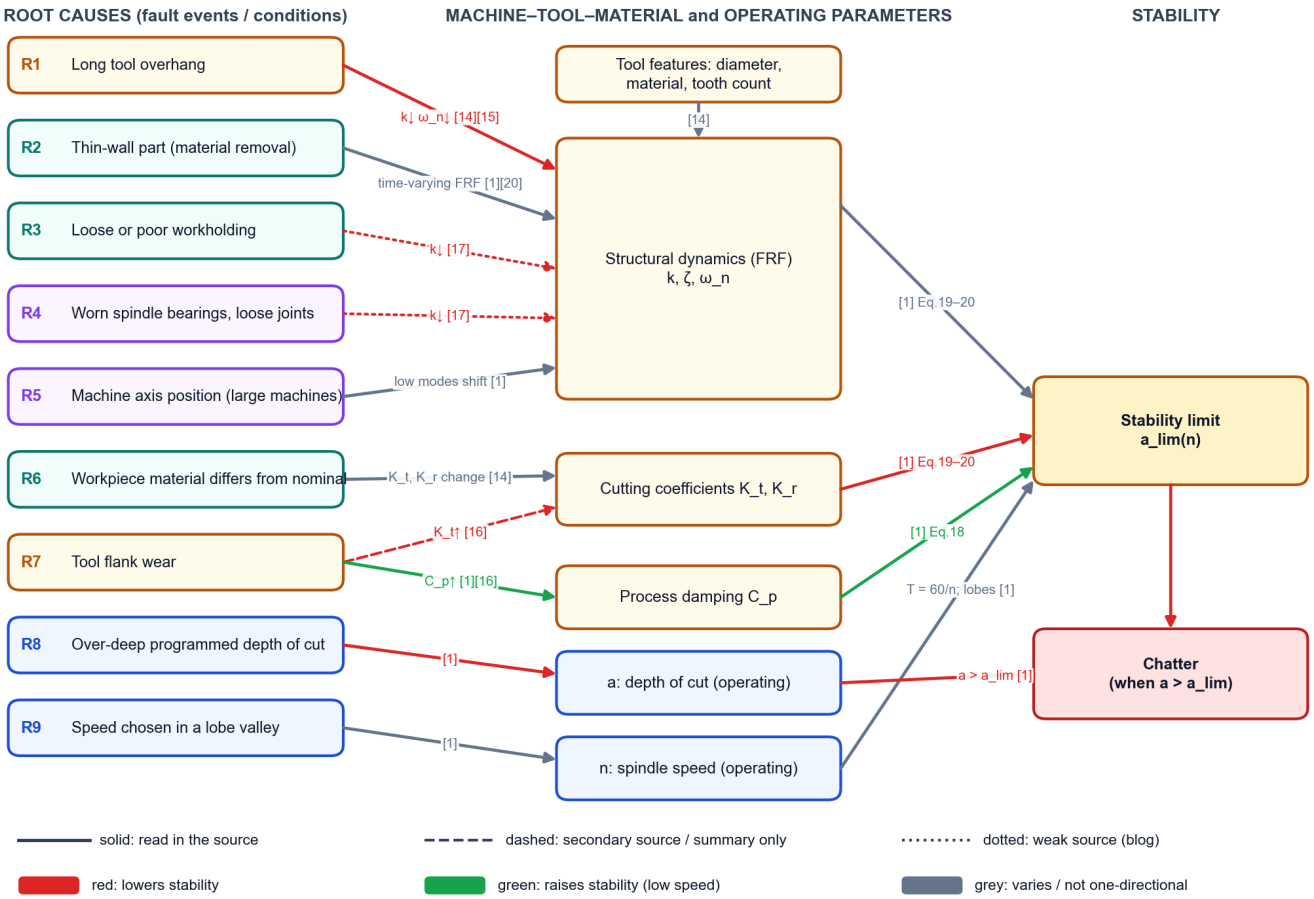


Figure 2. *Relations among root causes, machine–tool–material parameters, operating parameters and stability, with the source of each relation (bracketed numbers refer to the reference list).*

Table 3. *Root causes, affected parameters and the source of each relation.*

Group	Root cause	Affected parameters and direction	Source	Verification level
Tool – holder	R1 Long tool overhang	FRF: $k\downarrow$, $\omega_n\downarrow$; damping ζ: inconsistent across sources	[14][15]	read (2 milling papers)
Workpiece	R2 Thin-wall part	workpiece FRF varies with tool position as material is removed	[1][20]	[1] read; [20] summary only
Workpiece	R3 Loose or poor workholding	$k\downarrow$ on the workpiece side	[17]	engineering blog only
Machine	R4 Worn spindle bearings, loose joints	dynamic stiffness $k\downarrow$	[17] ([18])	blog; [18] only lists it as a failure mode, not about stability
Machine	R5 Machine axis position (large machines)	low-frequency modes shift with position	[1]	read
Workpiece	R6 Material differs from nominal	K_t, K_r change (e.g. $K_t \approx 774$–984 N/mm² for 6061 aluminium; ≈ 1913–2431 for 45 steel)	[14]	read
Tool – holder	R7 Tool flank wear	$K_t\uparrow$ (cutting force rises) [16]; $C_p\uparrow$ (process damping) [1][16][21]. **Net effect is not monotonic**	[1][16][21]	[1], [16] read; [21] summary only
Operation	R8 Over-deep programmed depth	a rises; chatter when $a > a_{lim}$	[1]	read
Operation	R9 Speed in a lobe valley	n changes the delay $T = 60/n$; tooth-passing frequency coincides with a mode	[1]	read

In addition, tool features (diameter, material, tooth count) also change the tool-tip FRF [14]; this is a design choice rather than a fault, so I draw it as a parameter and not as a root cause.

Comparison with the project's experiments. The "5 root causes" of Experiment 2 cover only part of the graph: "stiffness drop" mixes R1, R3, R4; "damping loss" is only part of R1; "tool wear" models only the $K_t\uparrow$ part of R7 and **ignores the stabilising $C_p\uparrow$**; "depth overload" is R8 and "RPM mismatch" is R9. Three causes, R2, R5 and R6, appear in neither experiment.

4. Experiment 1 – Physics simulation and Bayesian-network diagnosis

Goal: test whether the physics of chatter leaves distinguishable "fingerprints" and whether the hidden cause can be diagnosed from observable symptoms. **Method:** (i) compute the stability-lobe diagram analytically; (ii) simulate the time-domain DDE for 6 scenarios; (iii) derive 6 yes/no symptoms from signal features; (iv) infer the cause with a small Bayesian network. **Measured:** diagnosis accuracy (Top-1, Top-3) over 120 noisy runs.

4.1 Stability-lobe diagram: which speed and depth are safe?

A **stability-lobe** diagram shows, for each spindle speed, the deepest cut at which the machine still runs smoothly. **Below the curve** cutting is stable; **above it** chatter occurs. At certain speeds (lobe peaks) a much deeper cut is possible, while in the "valleys" only very shallow cuts are safe. Figure 3 is computed analytically with the project's Python code (*chatter_simulation.py*), reproducing two textbook examples from Altintas.

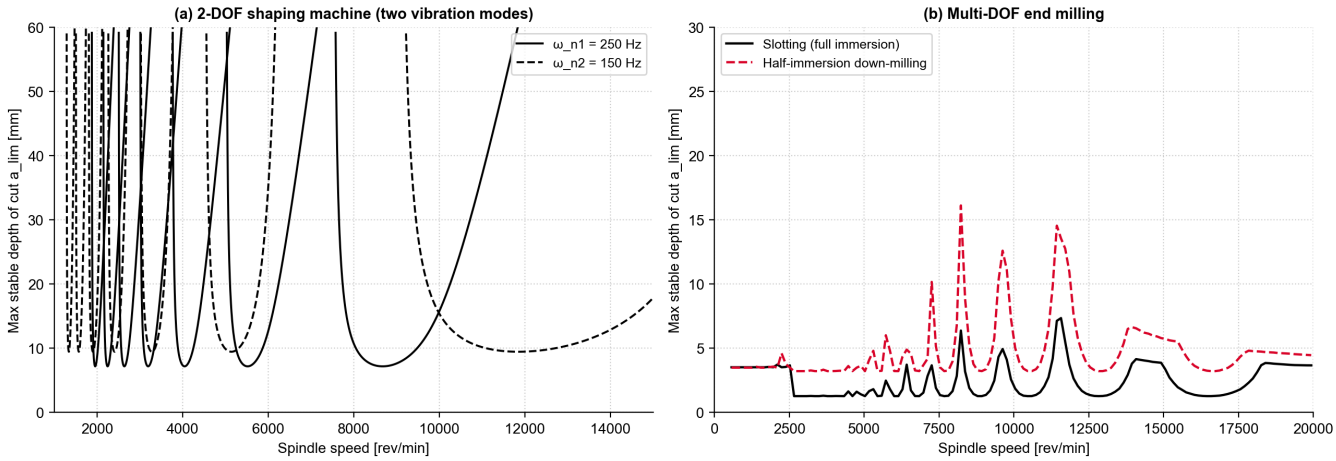


Figure 3. *Stability-lobe diagrams: (a) a 2-DOF shaping machine with two vibration modes at 250 Hz and 150 Hz; (b) slotting and half-immersion down-milling. Below a curve: stable cutting; above it: chatter.*

4.2 Vibration over time

The project has a **time-domain** simulator (*rca_chatter_bayesian.py*) that integrates the DDE numerically (Euler–Cromer): the force at time *t* depends on the waviness left by the previous pass (*T* = 60/rpm earlier). Figure 4 shows the waveform and spectrum of three cases: normal, stiffness −50% (the vibration frequency drops from 250 Hz to a measured ≈ 187 Hz) and excessive depth (vibration grows, i.e. chatter).

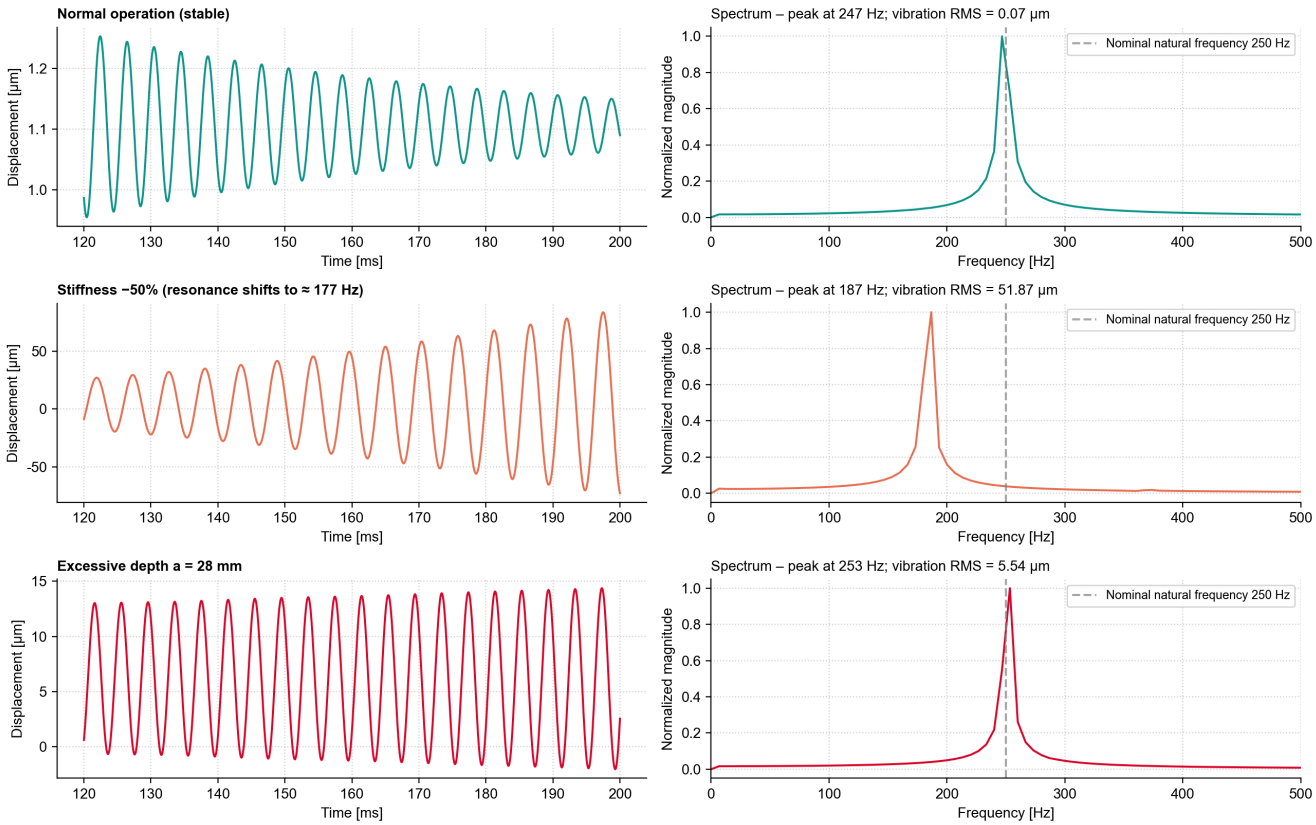


Figure 4. *Physical "fingerprint" of each cause: displacement waveform (left) and frequency spectrum (right). Vertical scales differ between rows.*

4.3 Diagnosis with a small Bayesian network

From each simulation we extract 6 yes/no "symptoms" (strong vibration? chatter peak? frequency shift? high force? large programmed depth? speed in a valley?) and use a **Bayesian network** with 6 hypotheses (normal + 5 causes) to compute each hypothesis' probability by Bayes' rule. Table 4 shows the results for 6 scenarios.

Table 4. *Bayesian-network diagnosis for 6 simulated scenarios.*

Scenario (true cause)	Vib. RMS (μm)	Peak freq. (Hz)	Diagnosis	Prob.	Correct?
Normal	0.07	247	Normal	99%	Yes
Stiffness −50%	51.87	187	Stiffness drop	97%	Yes
Damping loss (a = 8 mm)	2.55	253	Normal	99%	**No**
Tool wear (K_f +120%)	0.27	247	Tool wear	89%	Yes
Excessive depth (a = 28 mm)	5.54	253	Excessive depth	99%	Yes
Spindle 8,800 rpm	39.43	253	Unstable speed	91%	Yes

The “damping loss” scenario is misdiagnosed: at a = 8 mm, 3,800 rpm the simulation actually stays **stable** (vibration only 2.55 μm, below the 5 μm threshold), so every symptom is “no” and the network concludes “normal” – reasonable given the data, yet the fault label says damping loss. Over 120 additional noisy runs (20 × 6 scenarios): **Top-1 = 76.7%**, **Top-3 = 100%**.

4.4 Takeaways from Experiment 1

- Each cause leaves a different mark in the waveform and spectrum (frequency shifts down when stiffness drops; vibration grows when cutting too deep), so in principle they can be told apart from sensors.
- With hidden causes and observable symptoms as evidence (as on a real machine), the Bayesian network reaches Top-1 77% and Top-3 100%.
- Limits: only 6 hypotheses, hand-set probability tables and thresholds; the “damping loss” case shows a fault may leave no symptom if the operating point stays stable (see Section 3.6).

5. Experiment 2 – RCA benchmark with 14 variables

Experiment 1 has only 6 hypotheses and binary symptoms. Experiment 2 has more continuous variables: the algorithm sees **14 quantities** and must rank the true root cause among all 14. To run thousands of trials it uses a **simple algebraic model** instead of the time-domain solver (code: *run_full_paper_benchmark.py*). **Experiment 2 data does not come from the Experiment 1 simulation.** As discussed in Section 3.3, 5 of the 14 columns (k, ζ, K_t, a_{lim}, chatter severity) **cannot be measured on a real machine** yet are included in the observed data.

5.1 Simulated data and fault injection

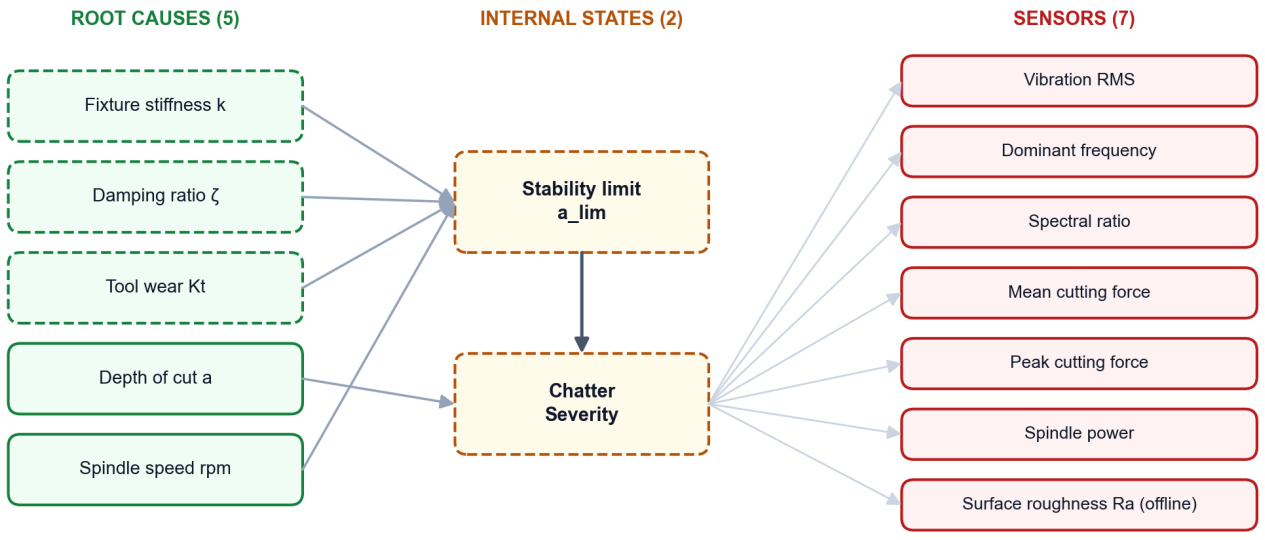


Figure 5. *The 14 variables and their direction of influence (arrows run from cause to effect). Dashed outline: not directly measurable on a real machine.*

Inputs – 5 **normalised** numbers (no SI units): *k*, ζ, K_t around 1.0 (3.5% spread); *a*, *rpm* around 0.50 (0.025 spread). So "rpm = 0.5" is **not** 0.5 revolutions per minute. **Outputs** – an N × 14 table computed as follows (noise = small random numbers imitating sensor error):

Col.	Quantity	How it is computed (simplified)
0–4	k, ζ, K_t, a, rpm	the 5 inputs plus sensor noise
5	a_{lim} – stability limit	$0.85 \cdot k \cdot \zeta / (K_t \cdot \text{material hardness}) \times (1 - 0.35 \cdot \sin(4\pi \cdot \text{rpm})) + \text{noise}$
6	Chatter Severity	$0.25 \cdot \ln(1 + e^{9 \cdot (a - a_{lim})}) + \text{noise}$; rises fast once a exceeds a_{lim}
7	Vibration RMS (µm)	0.12 + 1.5 × severity + noise
8	Dominant frequency (Hz)	$250 \cdot \sqrt{k} + 5 \times \text{severity} + \text{noise}$
9	Spectral ratio	saturating function of severity (0.08 → ≈ 0.78)
10	Mean cutting force (N)	300 × K_t × material hardness × a + noise
11	Peak cutting force (N)	mean force × (1.15 + 1.5 × severity) + noise
12	Spindle power (W)	0.75 × mean force × rpm + 0.12 × peak force + noise
13	Surface roughness Ra (µm)	0.35 + 2.5 × vibration RMS + 0.0008 × peak force + noise

Workpiece hardness varies by ±5% as an additional confounder.

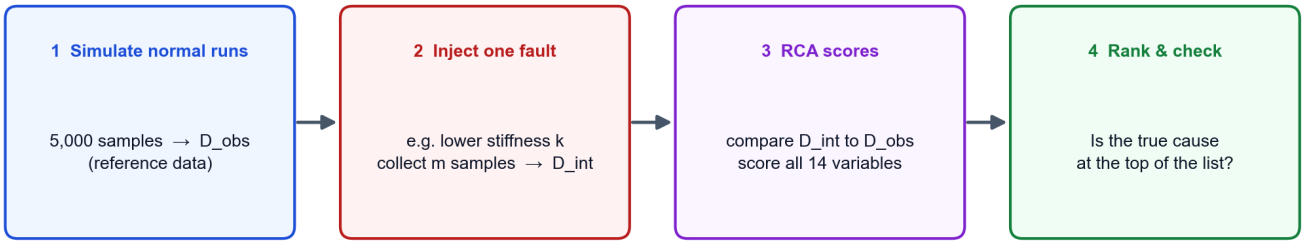


Figure 6. *Experimental procedure. "Normal" data is the reference; "anomalous" data has exactly one root cause shifted.*

How are faults injected? Each fault shifts one parameter by about 1.2–2.7 natural standard deviations (e.g. k −5.5%, tool wear raises K_t by ≈ 6.5%) – fairly subtle and easily masked by noise. Each trial has 5,000 normal samples (D_obs) and m anomalous samples (D_int, m from 5 to 100).

5.2 Algorithms and scoring

Scoring: each algorithm ranks all 14 variables from most to least suspect. **Top-1** = the true cause is ranked first; **Top-3 / Top-5** = it is within the first 3 / 5 (an engineer only needs to inspect a few items).

The six algorithms (how each variable is scored, as coded in the benchmark):

Name	Scoring rule
SimpleRCA	How far the largest anomalous value exceeds the 95th percentile of normal data. No causal graph.
BARO	Mean deviation from the median divided by the interquartile range (IQR) – a "robust" deviation. No causal graph.
RCD	After removing the effect of its parent variables, test whether the variable still correlates with "this sample is anomalous" (statistical test).
RCG	After subtracting what the parents explain, how many standard deviations the residual has shifted.
SmoothTraversal	Deviation of the variable minus 70% of its parent's deviation: if the parent already deviates a lot, the variable is less suspect.
BRCD	Similar "residual shift" idea, multiplied by a factor for the number of downstream variables, then turned into probabilities (softmax) over the 5 causes.

The bottom four algorithms use the causal graph in Figure 5. Idea by example: roughness rises sharply during chatter but only because vibration rose; the part of roughness that vibration does not explain stays normal → not suspect. SimpleRCA and BARO just see "roughness deviates a lot" and easily confuse symptoms with causes.

5.3 Results

Each value is the mean of 500 trials (100 × 5 fault types) with the standard error (after ±). A value of 0.71 means "right 71 times out of 100". **Remember that this is with the unmeasurable columns still included in the data** (Section 3.3).

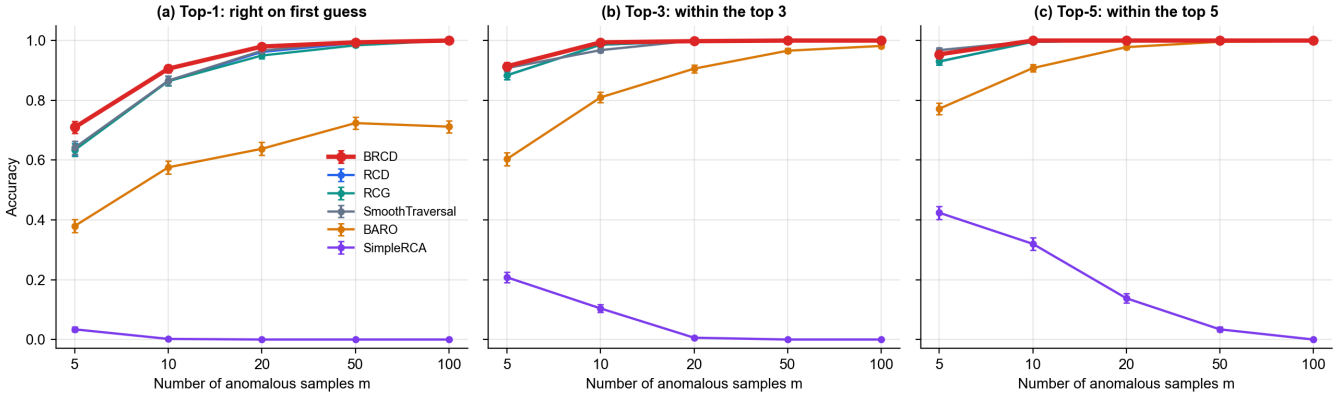


Figure 7. *Accuracy versus number of anomalous samples m. With more evidence all causal methods improve and converge to ≈ 100%; BARO plateaus near 0.7 (Top-1); SimpleRCA stays near 0 on Top-1.*

Table 5. *Top-1 by number of samples m.*

Algorithm	m = 5	m = 10	m = 20	m = 50	m = 100
BRCD	**0.71 ± 0.02**	**0.91 ± 0.01**	**0.98 ± 0.01**	**0.99 ± 0.00**	**1.00 ± 0.00**
RCD	0.64 ± 0.02	0.87 ± 0.02	0.96 ± 0.01	0.99 ± 0.00	1.00 ± 0.00
RCG	0.63 ± 0.02	0.86 ± 0.02	0.95 ± 0.01	0.98 ± 0.01	1.00 ± 0.00
SmoothTraversal	0.64 ± 0.02	0.87 ± 0.02	0.97 ± 0.01	1.00 ± 0.00	1.00 ± 0.00
BARO	0.38 ± 0.02	0.58 ± 0.02	0.64 ± 0.02	0.72 ± 0.02	0.71 ± 0.02
SimpleRCA	0.03 ± 0.01	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00

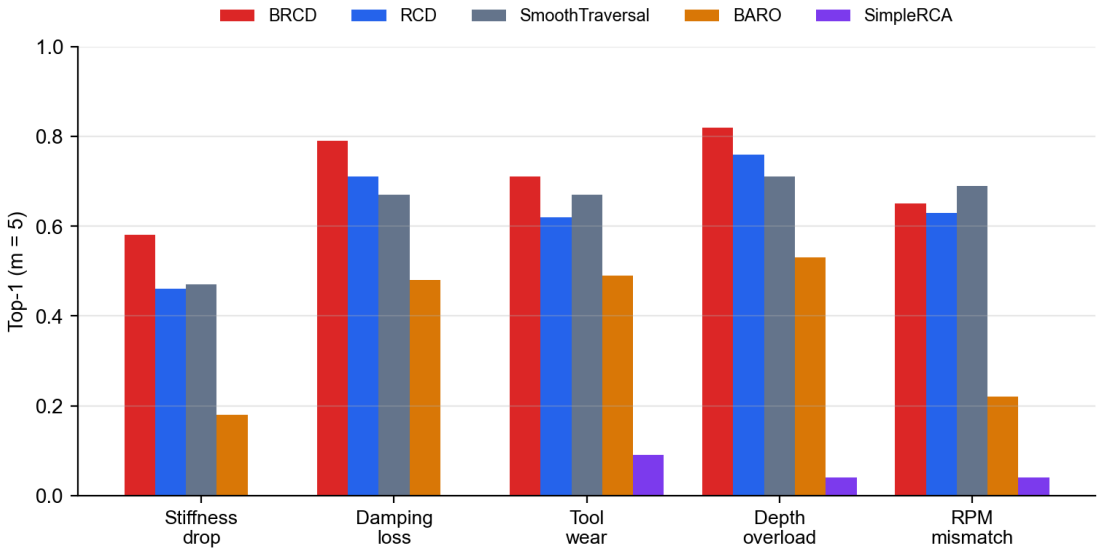


Figure 8. *Top-1 per fault type with only m = 5 samples: "stiffness drop" is hardest; "depth overload" and "damping loss" are easiest.*

Table 6. *The three scoring criteria at m = 5.*

Algorithm	Top-1	Top-3	Top-5
BRCD	0.71 ± 0.02	0.91 ± 0.01	0.95 ± 0.01
RCD	0.64 ± 0.02	0.88 ± 0.01	0.93 ± 0.01
RCG	0.63 ± 0.02	0.88 ± 0.01	0.93 ± 0.01
SmoothTraversal	0.64 ± 0.02	0.91 ± 0.01	0.97 ± 0.01
BARO	0.38 ± 0.02	0.60 ± 0.02	0.77 ± 0.02
SimpleRCA	0.03 ± 0.01	0.21 ± 0.02	0.42 ± 0.02

Reading the results:

- With little evidence (m = 5) BRCD beats the other causal methods by ≈ 7–8 percentage points of Top-1 (0.71 vs 0.63–0.64) – a small gap, a few standard errors wide; at m = 10 it is 4–5 points and from m = 50 there is almost no difference.
- If an engineer inspects the first 3 items (Top-3), the causal methods already contain the true cause ≈ 90% of the time with only 5 samples.
- The two methods without a causal graph do much worse – most likely because the symptoms (peak force, roughness… with large fluctuations) dominate; SimpleRCA even gets worse with more samples because the maximum value is more likely to land on a high-noise variable.

5.4 How does BRCD's probability change as samples arrive?

Figure 9 plots the probability BRCD assigns to the true cause as anomalous samples arrive one by one (t = 1…40; line = mean, shaded = 25–75% range over 60 runs). This is the **raw**, unsmoothed probability. After 40 samples the mean probability is only about **0.52** (guessing is 0.2) and the entropy is still ≈ **1.32** nats (maximum 1.61): the probability rises but does **not** approach 1, and uncertainty falls very little. The cause is the softmax in the code with a fixed factor of 0.40, so this number should be read as a **ranking score**, not a calibrated probability. (The convergence plot in the earlier version of the benchmark blended in the prior and added noise, so it looked nicer; this report does not use it.)

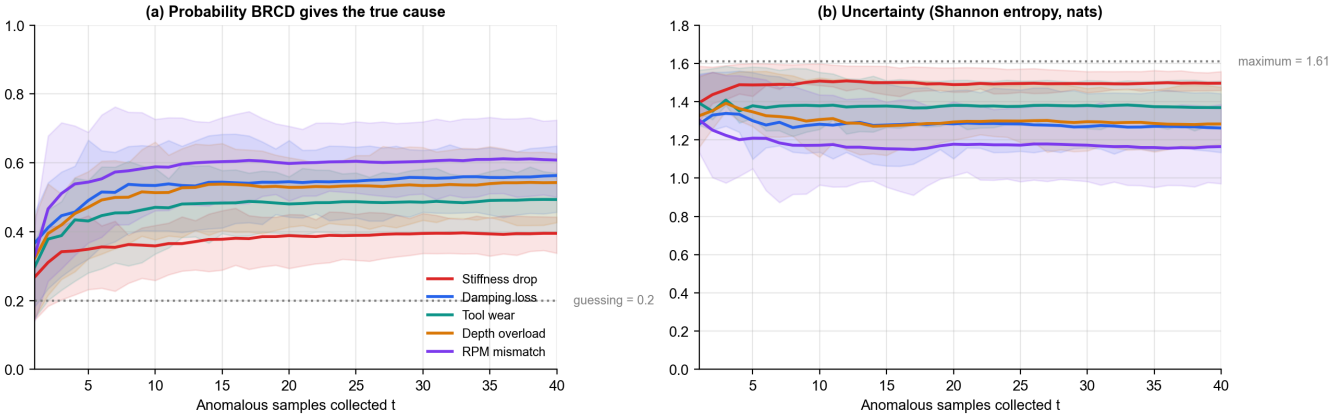


Figure 9. *BRCD convergence versus sample count t: (a) probability assigned to the true cause; (b) entropy of the distribution over the 5 causes.*

6. Discussion and limitations

6.1 Comparing the two experiments

The two experiments answer different questions and complement each other, but they are **not yet connected**: Experiment 2's data is not generated by Experiment 1's solver.

Aspect	Experiment 1: physics + Bayesian network	Experiment 2: 14-variable benchmark
Physics model	single-mode DDE, numerical integration; analytic stability lobes	simple algebraic formulas, no time-domain solver
Data given to RCA	6 binary symptoms derived from sensor features	an N × 14 table of continuous variables
Causes	5 causes + “normal”, **hidden variables**	5 parameters, but their columns **are in the data**
Method	Bayesian network, hand-set probability tables	6 algorithms (BRCD, RCD, RCG, SmoothTraversal, BARO, SimpleRCA), simplified versions
Scale	6 scenarios + 120 noisy runs	2,500 trials per algorithm
Main result	Top-1 76.7%, Top-3 100%	BRCD Top-1 71% ($m = 5$) → ≈ 100% ($m \geq 50$)
Strength	follows the observability principle: hidden causes, symptoms as evidence	large scale; compares many algorithms and sample sizes
Weakness	few hypotheses; hand-set tables and thresholds	exposes unmeasurable variables; scored on a model we wrote

The two results can be read as follows: Experiment 1 shows that diagnosis from observable symptoms is feasible in a small setting; Experiment 2 shows that causal algorithms clearly beat a simple statistical method, but the gap among causal algorithms is small and the absolute numbers are inflated by the exposure of unmeasurable variables.

6.2 Limitations

- **The data is simulated, not from a real machine.** The formulas are the project’s own and the algorithms are scored on that same model. 100% does not mean 100% on a real CNC machine.
- **The Experiment 2 benchmark exposes unmeasurable variables.** k, ζ, K_t, a_{lim} and chatter severity are in the observed data (Section 3.3). When a cause is faulted its own column shifts, so the task is far easier than inferring hidden causes from real sensors. No experiment in this report measures accuracy when **only** observable quantities are used; the numbers would be lower, but by how much is unknown.
- **Not modelled:** closed-loop control (Section 3.4), bistable zones and other bifurcations (Section 3.5), process damping and the dual effect of tool wear, time-varying dynamics, and the many-to-many mapping between faults and parameters (Sections 3.6, 3.8).
- **Experiments 1 and 2 are separate.** The 14-variable benchmark uses algebraic formulas, not Experiment 1’s time-domain solver. The Experiment 1 Bayesian network has hand-set probability tables and symptom thresholds, and the “large programmed depth” symptom is read directly from the known value of a.
- **The RCA algorithms are simplified re-implementations** (e.g. BRCD here is a linear-Gaussian residual-shift score followed by a softmax), **not** the original code of the papers, so this is not a reproduction of the ICML 2026 results. The +0.12 bonus per downstream variable is the project’s own design choice.
- **The causal algorithms are given the correct causal graph**, and the two datasets have different noise levels (factor 1.0 vs 1.1), which may leak a little irrelevant “hint”.

7. Next steps: Bayesian causal network + Knowledge Graph + RAG

Combining a causal Bayesian network (CBN), a knowledge graph (KG) and RAG for RCA is **feasible and has precedent** in other domains: Wehner et al. [7] combine a CBN with a KG for RCA in electric-vehicle manufacturing (the KG shortens learning time and reduces spurious causal links); Ouarrad et al. [8] use a KG with RAG and an LLM as the decision and explanation layer for RCA in automotive HiL systems, reaching Top-1 of 90% and 94% on two case studies. I have not found work applying it directly to chatter, and results from other domains do not transfer automatically.

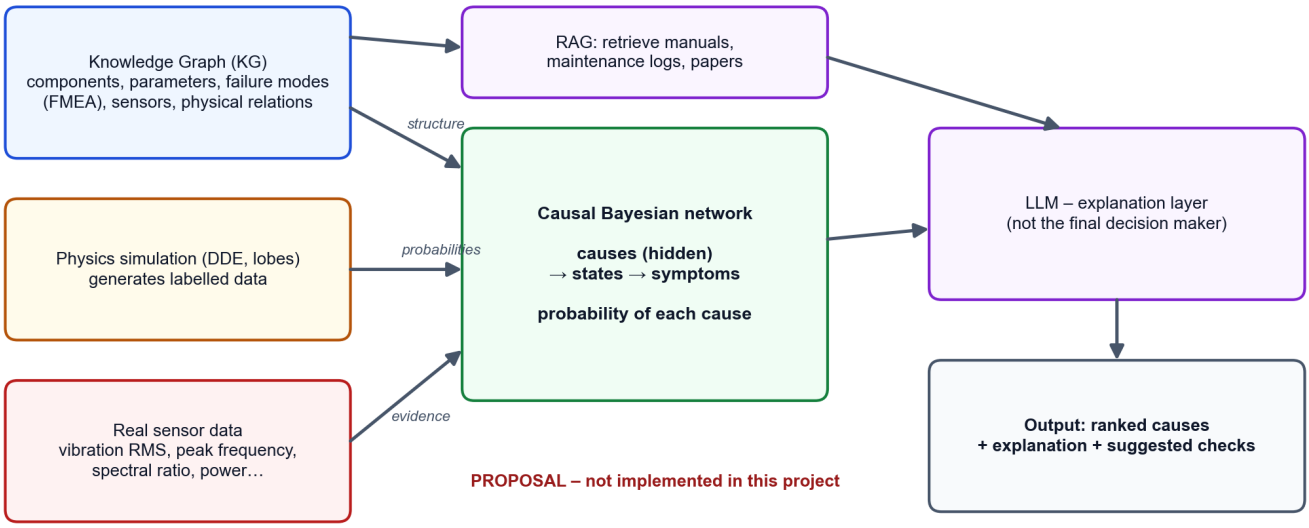


Figure 10. *Proposed architecture. The KG provides structure, the physics simulation provides probabilities and real sensors provide evidence for the Bayesian network; RAG and the LLM only explain and suggest.*

Division of roles:

- **KG:** components, parameters, failure modes (from FMEA), sensors and physical relations (such as the formulas in Section 3.2); constrains the structure of the Bayesian network.
- **Causal Bayesian network:** quantitative inference; causes are **hidden variables** and evidence is only observable features. Probabilities can be learned from Experiment 1's DDE solver.
- **RAG:** retrieve machine manuals, maintenance logs and papers to explain and suggest checks. The **LLM** should only be an explanation layer, not the final decision maker.

Suggested roadmap:

- Fix the Experiment 2 benchmark: **hide** the unmeasurable columns (k, ζ, K_t, a_{lim}, chatter severity) and re-measure accuracy.
- Generate benchmark data from the DDE solver (Experiment 1) or stability lobes instead of the algebraic formulas; add process damping and bistable conditions.
- Use the Section 3.8 graph as the starting structure: separate the two cause levels (fault event and parameter), model tool wear with both K_t and C_p, add R2, R5, R6; try a learned or partly wrong graph to test robustness.
- Build the KG from FMEA and physical relations, then add RAG/LLM as an explanation layer; evaluate the correctness of the explanations separately.
- Run the original code of RCD, BARO, BRCD instead of simplified versions; calibrate BRCD's probabilities; test on real data with accelerometers, spindle current and cutting force.

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